

ARINC 717 DATAFRAME LAYOUT DOCUMENT – MAPPING

Introduction

THE CURRENT DOCUMENT IS DESIGNED TO CAPTURE THE PROPERTIES OF THE RECORDED PARAMETERS BY THE FDR. THIS TYPE OF DOCUMENT IS ALSO KNOWN AS INTERFACE CONTROL DOCUMENT (ICD), OR FLIGHT DATA RECORDER PARAMETER LIBRARY (FDRPL). THESE WILL BE USED IN CREATING WHAT IS COMMONLY KNOWN AS A DATAFRAME, OR DATABASE.

Aircraft Type

CN235-220

Version Control

Initial Issue 0 dated 15 Oct 2025

DFDAU Part Number

FDS81-2532 PT Indonesian Aerospace MAU

Customer Name

PT Dirgantara Indonesia

Other Details

This dataframe layout has been created using the ICD document, FDR Parameter List for CN235-220 GFA document no. 35/NAL4/CNC217/23/010 Rev C dated 11 Aug 2025 and the PTDI MAU document from FDS Engineering dated 18 Jun 2024.

Document Approval

Prepared by:	Charles Rono	_____	Date	_____
Checked by:		_____	Date	_____
Approved by:		_____	Date	_____

RECORD OF REVISIONS



03 DEC 2025

280 Bath Road, UB7 0DQ, UK
CAA.UK.145.01332

REVISION	DATE	CHANGE DESCRIPTION	PREPARED / REVISED BY
0	15 Oct 2025	Initial Issue	Charles Rono



03 DEC 2025

260 Bath Road, UB7 0DQ, UK
CAA.UK, 145.01332

Dataframe Layout Document – Recorded Parameters



Dataframe WPS : 256
Superframe Present: No
Revision No: Initial Issue 0
Issue Date: 15/10/2025

Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
28VDC Power Input	BNR	1	247	12	1	0 (all 4)	-	3.5, 0.01	-	-	
AOAL LH Angle Of Attack	BNR	0.5	14-15 , 142-143	9-1	12-9	0	-	0.044	-	-	
AOAR RH Angle Of Attack	BNR	0.5	16-17 , 144-145	9-1	12-9	0	-	0.044	-	-	
AP Armed Mode Alt Mode Armed	Discrete	1	178	4	4	0	-	1	Alt Mode Armed	-	
AP Armed Mode APR Mode Armed	Discrete	1	178	10	10	0	-	1	APR Mode Armed	-	
AP Armed Mode BC Mode Armed	Discrete	1	178	9	9	0	-	1	BC Mode Armed	-	
AP Armed Mode GS Mode Armed	Discrete	1	177	2	2	0	-	1	GS Mode Armed	-	
AP Armed Mode LNAV Mode Armed	Discrete	1	178	6	6	0	-	1	LNAV Mode Armed	-	
AP Armed Mode LOC Armed	Discrete	1	178	8	8	0	-	1	LOC Armed	-	
AP Armed Mode Nav Mode Armed	Discrete	1	178	5	5	0	-	1	Nav Mode Armed	-	
AP Armed Mode VNAV Mode Armed	Discrete	1	178	7	7	0	-	1	VNAV Mode Armed	-	
AP Armed Mode VOR Armed	Discrete	1	177	3	3	0	-	1	VOR Armed	-	
AP Lateral Mode Apr Mode Active	Discrete	1	181	7	7	0	-	1	Apr Mode Active	-	
AP Lateral Mode BC Sub-Mode Active	Discrete	1	182	6	6	0	-	1	BC Sub- Mode Active	-	
AP Lateral Mode Capture Sub-Mode Active	Discrete	1	181	1	1	0	-	1	Capture Sub-Mode Active	-	
AP Lateral Mode CSW Mode Active	Discrete	1	181	9	9	0	-	1	CSW Mode Active	-	
AP Lateral Mode GO Around Mode Active	Discrete	1	182	11	11	0	-	1	GO Around	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
									Mode Active		
AP Lateral Mode Half Bank Active	Discrete	1	182	5	5	0	-	1	Half Bank Active	-	
AP Lateral Mode Hdg Mode Active	Discrete	1	182	4	4	0	-	1	Hdg Mode Active	-	
AP Lateral Mode Intercept Sub-Mode	Discrete	1	181	5	5	0	-	1	Intercept Sub-Mode	-	
AP Lateral Mode LNAV Sub-Mode Active	Discrete	1	182	3	3	0	-	1	LNAV Sub-Mode Active	-	
AP Lateral Mode LOC Sub-Mode Active	Discrete	1	182	7	7	0	-	1	LOC Sub-Mode Active	-	
AP Lateral Mode Nav Mode Active	Discrete	1	181	6	6	0	-	1	Nav Mode Active	-	
AP Lateral Mode Roll Mode Active	Discrete	1	182	12	12	0	-	1	Roll Mode Active	-	
AP Lateral Mode Track Sub-Mode	Discrete	1	181	3	3	0	-	1	Track Sub-Mode	-	
AP Lateral Mode VOR Sub-Mode Active	Discrete	1	181	2	2	0	-	1	VOR Sub-Mode Active	-	
AP Vertical Mode ALT Hold Mode Active	Discrete	1	180	5	5	0	-	1	ALT Hold Mode Active	-	
AP Vertical Mode Capture Sub-Mode Active	Discrete	1	179	9	9	0	-	1	Capture Sub-Mode Active	-	
AP Vertical Mode CWS Mode Active	Discrete	1	179	5	5	0	-	1	CWS Mode Active	-	
AP Vertical Mode GO Around Mode Active	Discrete	1	180	8	8	0	-	1	GO Around Mode Active	-	
AP Vertical Mode GS Mode Active	Discrete	1	180	11	11	0	-	1	GS Mode Active	-	
AP Vertical Mode IAS Mode Active	Discrete	1	180	3	3	0	-	1	IAS Mode Active	-	
AP Vertical Mode Pitch Mode Active	Discrete	1	179	8	8	0	-	1	Pitch Mode Active	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
AP Vertical Mode TOGA Around Mode Active	Discrete	1	180	7	7	0	-	1	TOGA Around Mode Active	-	
AP Vertical Mode VNAV Mode Active	Discrete	1	180	4	4	0	-	1	VNAV Mode Active	-	
AP Vertical Mode VS Mode Active	Discrete	1	180	6	6	0	-	1	VS Mode Active	-	
Baro Set (inHg) Baro Set (inHg)	BCD	1	154-155	9-1	12-7	0	-	1	-	-	
Baro Set (mbar) Baro Set (mbar)	BCD	1	152-153	9-1	12-7	0	-	1	-	-	
Brake Pressure Brake Pressure Emergency	BNR	1	96-97	9-1	12-9	0	-	1	-	-	
Brake Pressure Brake Pressure Normal	BNR	1	94-95	9-1	12-9	0	-	1	-	-	
Caution Messages Aux Pitot Heater	Discrete	1	186	4	4	0	-	1	Aux Pitot Heater	-	
Caution Messages Hyd Oil Hot Caution	Discrete	1	184	11	11	0	-	1	Hyd Oil Hot Caution	-	
Caution Messages Ice Form Caution	Discrete	1	184	4	4	0	-	1	Ice Form Caution	-	
Caution Messages LH AC Gen Caution	Discrete	1	184	8	8	0	-	1	LH AC Gen Caution	-	
Caution Messages LH Beta Light	Discrete	1	184	3	3	0	-	1	LH Beta Light	-	
Caution Messages LH Hyd Pressure Caution	Discrete	1	184	10	10	0	-	1	LH Hyd Pressure Caution	-	
Caution Messages LH Inlet Ice Caution	Discrete	1	184	6	6	0	-	1	LH Inlet Ice Caution	-	
Caution Messages LH Pitot Heater Caution	Discrete	1	184	5	5	0	-	1	LH Pitot Heater Caution	-	
Caution Messages LH TRU Caution	Discrete	1	184	9	9	0	-	1	LH TRU Caution	-	
Caution Messages LH Wshld Heat Caution	Discrete	1	184	7	7	0	-	1	LH Wshld Heat Caution	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
Caution Messages RH AC Gen Caution	Discrete	1	186	8	8	0	-	1	RH AC Gen Caution	-	
Caution Messages RH Beta Light	Discrete	1	186	3	3	0	-	1	RH Beta Light	-	
Caution Messages RH Hyd Pressure Caution	Discrete	1	186	10	10	0	-	1	RH Hyd Pressure Caution	-	
Caution Messages RH Inlet Ice Caution	Discrete	1	186	6	6	0	-	1	RH Inlet Ice Caution	-	
Caution Messages RH Pitot Heater Caution	Discrete	1	186	5	5	0	-	1	RH Pitot Heater Caution	-	
Caution Messages RH TRU Caution	Discrete	1	186	9	9	0	-	1	RH TRU Caution	-	
Caution Messages RH Wshld Heat Caution	Discrete	1	186	7	7	0	-	1	RH Wshld Heat Caution	-	
COMM1 Communication Event 1	Discrete	1	248	6	6	0	-	1	Open	Ground	
COMM2 Communication Event 2	Discrete	1	248	7	7	0	-	1	Open	Ground	
COMM3 Communication Event 3	Discrete	1	248	8	8	0	-	1	Open	Ground	
COMM4 Communication Event 4	Discrete	1	249	1	1	0	-	1	Open	Ground	
COMM5 Communication Event 5	Discrete	1	249	2	2	0	-	1	Open	Ground	
COMM6 Communication Event 6	Discrete	1	249	3	3	0	-	1	Open	Ground	
COMM7 Communication Event 7	Discrete	1	249	4	4	0	-	1	Open	Ground	
COMM8 Communication Event 8	Discrete	1	249	5	5	0	-	1	Open	Ground	
Computer Fail ADC #1 Failed	Discrete	1	213	5	5	0	-	1	ADC #1 Failed	-	
Computer Fail ADC #2 Failed	Discrete	1	213	6	6	0	-	1	ADC #2 Failed	-	
Computer Fail ADF #1 Failed	Discrete	1	212	5	5	0	-	1	ADF #1 Failed	-	
Computer Fail ADF #2 Failed	Discrete	1	212	6	6	0	-	1	ADF #2 Failed	-	



Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
Computer Fail AHRS #1 Failed	Discrete	1	213	3	3	0	-	1	AHRS #1 Failed	-	
Computer Fail AHRS #2 Failed	Discrete	1	213	4	4	0	-	1	AHRS #2 Failed	-	
Computer Fail AP Failed	Discrete	1	212	8	8	0	-	1	AP Failed	-	
Computer Fail DAU #1A Failed	Discrete	1	212	4	4	0	-	1	DAU #1A Failed	-	
Computer Fail DAU #1B Failed	Discrete	1	212	5	5	0	-	1	DAU #1B Failed	-	
Computer Fail DAU #2A Failed	Discrete	1	212	6	6	0	-	1	DAU #2A Failed	-	
Computer Fail DAU #2B Failed	Discrete	1	212	7	7	0	-	1	DAU #2B Failed	-	
Computer Fail DME Failed	Discrete	1	212	1	1	0	-	1	DME Failed	-	
Computer Fail GPS #1 Failed	Discrete	1	213	7	7	0	-	1	GPS #1 Failed	-	
Computer Fail GPS #2 Failed	Discrete	1	213	8	8	0	-	1	GPS #2 Failed	-	
Computer Fail Same_Nav	Discrete	1	213	9	9	0	-	1	Same_Na v	-	
Computer Fail SAND #1 Failed	Discrete	1	210	7	7	0	-	1	SAND #1 Failed	-	
Computer Fail SAND #2 Failed	Discrete	1	210	8	8	0	-	1	SAND #2 Failed	-	
Computer Fail SDI (Parameter 51)	Discrete	1	213	2	1	0	-	1	0 – PFD #1 1 – MFD #1 2 – PFD #2 3 – MFD #2		
Computer Fail SDI (Parameter 52)	Discrete	1	215	2	1	0	-	1	0 – PFD #1 1 – MFD #1 2 – PFD #2 3 – MFD #2		
Computer Fail TCAS Failed	Discrete	1	212	2	2	0	-	1	TCAS Failed	-	
Computer Fail TDR Failed	Discrete	1	212	4	4	0	-	1	TDR Failed	-	
Computer Fail WXR Failed	Discrete	1	212	3	3	0	-	1	WXR Failed	-	
Date Date Day Ones Digit	BCD	1	165	7	4	0	day	1	-	-	
Date Date Day Tens Digit	BCD	1	165	9	8	0	day	1	-	-	
Date	BCD	1	165-166	2-1	12-11	0	month	1	-	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
Date Month Ones Digit											
Date	BCD	1	165	3	3	0	month	1	-	-	
Date Month Tens Digit											
Date	BCD	1	166	6	3	0	year	1	-	-	
Date Year Ones Digit											
Date	BCD	1	166	10	7	0	year	1	-	-	
Date Year Tens Digit											
Decision Height	BNR	1	107-108	8-1	12-8	0	feet	1	-	-	
Selected Decision Height											
DME Distance	BCD	1	113	10	9	0	Nm	-	-	-	
DME Distance #1 (Hundreds)											
DME Distance	BCD	1	114	8	5	0	Nm	-	-	-	
DME Distance #1 (Hundredths)											
DME Distance	BCD	1	113	8	5	0	Nm	-	-	-	
DME Distance #1 (Tens)											
DME Distance	BCD	1	114	12	9	0	Nm	-	-	-	
DME Distance #1 (Tenths)											
DME Distance	BCD	1	113	4	1	0	Nm	-	-	-	
DME Distance #1 (Units)											
DME Distance	BCD	1	115	10	9	0	Nm	-	-	-	
DME Distance #2 (Hundreds)											
DME Distance	BCD	1	116	8	5	0	Nm	-	-	-	
DME Distance #2 (Hundredths)											
DME Distance	BCD	1	115	8	5	0	Nm	-	-	-	
DME Distance #2 (Tens)											
DME Distance	BCD	1	116	12	9	0	Nm	-	-	-	
DME Distance #2 (Tenths)											
DME Distance	BCD	1	115	4	1	0	Nm	-	-	-	
DME Distance #2 (Units)											
DME Frequency	BCD	1	33	10	10	0	MHz	1	0.05	0.00	
DME Frequency #1 (If)											
DME Frequency	BCD	1	32	9	7	0	MHz	1	-	-	
DME Frequency #1 (Tens)											
DME Frequency	BCD	1	32-33	2-1	12-11	0	MHz	1	-	-	
DME Frequency #1 (Tenths)											
DME Frequency	BCD	1	32	6	3	0	MHz	1	-	-	
DME Frequency #1 (Units)											
DME Frequency	BCD	1	36	10	10	0	MHz	1	0.05	0.00	
DME Frequency #2 (If)											
DME Frequency	BCD	1	35	9	7	0	MHz	1	-	-	
DME Frequency #2 (Tens)											
DME Frequency	BCD	1	35-36	2-1	12-11	0	MHz	1	-	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
DME Frequency #2 (Tenths)											
DME Frequency DME Frequency #2 (Units)	BCD	1	35	6	3	0	MHz	1	-	-	
Drift Angle Drift Angle	BNR	1	237-238	9-1	12-9	0	deg	0.05	-	-	
Engine Oil Press LH Engine Oil Pressure	BNR	1	84-85	9-1	12-9	0	psi	0.1	-	-	
Engine Oil Press RH Engine Oil Pressure	BNR	1	86-87	9-1	12-9	0	psi	0.1	-	-	
Engine Oil Temp LH Engine Oil Temperature	BNR	1	63-64	9-1	12-9	0	deg C	0.1	-	-	
Engine Oil Temp RH Engine Oil Temperature	BNR	1	78-79	9-1	12-9	0	deg C	0.1	-	-	
FGC Status AP Engaged	Discrete	1	188	9	9	0	-	1	AP Engaged	-	
FGC Status FD Engaged	Discrete	1	188	10	10	0	-	1	FD Engaged	-	
FGC Status YD Engaged	Discrete	1	188	11	11	0	-	1	YD Engaged	-	
Flap Position Flap Position	BNR	1	101-102	8-1	12-8	0	deg	0.05	-	-	
FRAME Frame Counter	BNR	1	65	12	1	0	decimal counts	-	-	-	Raw Decimal
GMT (Internal RTC) GMT Date (Internal RTC) Days	BCD	1	24	9	4	0	days	1	-	-	
GMT (Internal RTC) GMT Date (Internal RTC) Months	BCD	1	24-25	3-1	12-11	0	months	1	-	-	
GMT (Internal RTC) GMT Date (Internal RTC) Year	BCD	1	25	10	3	0	year	1	-	-	
GMT (Internal RTC) GMT Time (Internal RTC) Hours	BNR	1	24	9	4	0	hour	-	-	-	Raw Decimal
GMT (Internal RTC) GMT Time (Internal RTC) Minutes	BNR	1	24-25	3-1	12-10	0	minutes	-	-	-	Raw Decimal
GMT (Internal RTC) GMT Time (Internal RTC) Seconds	BNR	1	25	9	4	0	seconds	-	-	-	Raw Decimal
Ground Speed Ground Speed	BNR	1	224-225	8-1	12-6	0	knots	0.125	-	-	
Heading Mag Heading #1	BNR	1	233-234	9-1	12-9	0	deg	0.0055	-	-	
Heading Mag Heading #2	BNR	1	235-236	9-1	12-9	0	deg	0.0055	-	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
HWCI HWCI Configuration Hash (SF1)	BNR	1	193	8	1	1	hex	1	-	-	Raw Hexadecimal
HWCI HWCI Configuration Hash (SF2)	BNR	1	193	8	1	2	hex	1	-	-	Raw Hexadecimal
HWCI HWCI Configuration Hash (SF3)	BNR	1	193	8	1	3	hex	1	-	-	Raw Hexadecimal
HWCI HWCI Configuration Hash (SF4)	BNR	1	193	8	1	4	hex	1	-	-	Raw Hexadecimal
Hydraulic Oil Qty Hydraulic Oil Qty	BNR	1	92-93	9-1	12-9	0	%	0.1	-	-	
Hydraulic Pressure Hydraulic Pressure	BNR	1	99-100	8-1	12-9	0	psi	1	-	-	
IAS Indicated Airspeed #1	BNR	1	121-122	9-1	12-7	0	knots	0.0625	-	-	
IAS Indicated Airspeed #2	BNR	1	123-124	9-1	12-7	0	knots	0.0625	-	-	
IBIT Current Data (EEPROM) Internal	BNR	1	22-23	12-1	12-1	0	-	-	-	-	Raw Hexadecimal
IBIT Result (Fault Code) 12V0 Internal Fault	Discrete	1	129	4	4	0	-	-	Fault	-	
IBIT Result (Fault Code) 28V0 Internal Fault	Discrete	1	129	8	8	0	-	-	Fault	-	
IBIT Result (Fault Code) 3V3 Internal Fault	Discrete	1	129	3	3	0	-	-	Fault	-	
IBIT Result (Fault Code) 5V0 Internal Fault	Discrete	1	129	5	5	0	-	-	Fault	-	
IBIT Result (Fault Code) A717 Loopback or 3V3 or 5V0 Internal Fault	Discrete	1	129	1	1	0	-	-	Fault	-	
IBIT Result (Fault Code) ARINC 717 Loopback Test Internal	Discrete	1	129	2	2	0	-	-	Fault	-	
IBIT Result (Fault Code) Primary Oscillator Fail Internal Fault	Discrete	1	129	9	9	0	-	-	Fault	-	
IBIT Result (Fault Code) RTC Internal Fault	Discrete	1	129	6	6	0	-	-	Fault	-	
IBIT Result (Fault Code) Secondary Oscillator Fail Internal Fault	Discrete	1	129	10	10	0	-	-	Fault	-	
IBIT Result (Fault Code) VMax Internal Fault	Discrete	1	129	7	7	0	-	-	Fault	-	
ILS Glidepath ILS Glidepath	BNR	1	111-112	9-1	12-9	0	ddm	0.0001	-	-	

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ILS Localizer ILS Localizer	BNR	1	109-110	9-1	12-9	0	ddm	0.0001	-	-	
ITT LH ITT	BNR	1	239-240	9-1	12-5	0	deg C	0.02	-	-	
ITT RH ITT	BNR	1	241-242	9-1	12-5	0	deg C	0.02	-	-	
Lateral Acc Lateral Acceleration	BNR	0.125	4, 68, 132, 196	12	1	0	G	-1.08, 0.002	-	-	
Latitude Latitude#1	BNR	1	55-56	9-1	3-12	0	deg	0.00017	-	-	
Latitude Latitude#2	BNR	1	57-58	9-1	3-12	0	deg	0.00017	-	-	
LH Aileron Pos LH Roll Axis Control Surface Pos	BNR	0.25	6, 70, 134, 198	12	1	0	deg	-23.527, 0.01	-	-	
LH Elevator Pos LH Pitch Control Surface/Elevator Pos	BNR	0.25	8, 72, 136, 200	12	1	0	deg	-40, 0.0195	-	-	
LHReverse Thrust Reverser Position LH	Discrete	1	248	4	4	0	-	1	Open	Ground	
Longitude Longitude#1	BNR	1	59-60	9-1	12-3	0	deg	0.00017	-	-	
Longitude Longitude#2	BNR	1	61-62	9-1	12-3	0	deg	0.00017	-	-	
Longitudinal Acc Longitudinal Acceleration	BNR	0.125	3, 67, 131, 195	12	1	0	G	-1.08, 0.002	-	-	
Marker Beacon Marker Beacon Inner	Discrete	1	151	3	3	0	-	1	Active	-	
Marker Beacon Marker Beacon Middle	Discrete	1	151	4	4	0	-	1	Active	-	
Marker Beacon Marker Beacon Outer	Discrete	1	151	5	5	0	-	1	Active	-	
MFD #1 Format - Map Display - Weather Radar Display - Traffic Display - HSI Display - Audio-Radio Display - Nav Log Display - Engine Display - Doors-Gear Display - Full Map Display - Full Engine Display	Discrete	1	191-192	8-1	12-5	0	-	1	0-0 1-1 2-2 3-3 4-4 5-5 6-6 7-7 8-8 9-9		Raw decimal and their corresponding output values have not been provided hence coded as shown.
MFD #1 Format Essential Mode	Discrete	1	192	2	2	0	-	1	Essential Mode	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
MFD #2 Format - Map Display - Weather Radar Display - Traffic Display - HSI Display - Audio-Radio Display - Nav Log Display - Engine Display - Doors-Gear Display - Full Map Display - Full Engine Display	Discrete	1	208-209	8-1	12-5	0	-	1	0-0 1-1 2-2 3-3 4-4 5-5 6-6 7-7 8-8 9-9	-	Raw decimal and their corresponding output values have not been provided hence coded as shown.
MFD #2 Format Essential Mode	Discrete	1	209	2	2	0	-	1	Essential Mode	-	
Navigation Source ADC #1 Source	Discrete	1	210	3	3	0	-	1	ADC #1 Source	-	
Navigation Source ADC #2 Source	Discrete	1	210	4	4	0	-	1	ADC #2 Source	-	
Navigation Source AHRS #1 Source	Discrete	1	210	1	1	0	-	1	AHRS #1 Source	-	
Navigation Source AHRS #2 Source	Discrete	1	210	2	2	0	-	1	AHRS #2 Source	-	
Navigation Source GPS #1 Source	Discrete	1	210	5	5	0	-	1	GPS #1 Source	-	
Navigation Source GPS #2 Source	Discrete	1	210	6	6	0	-	1	GPS #2 Source	-	
Navigation Source (ADF #1)	Discrete	1	211	6	6	0	-	1	Navigation Source (ADF #1)	-	
Navigation Source (ADF #2)	Discrete	1	211	7	7	0	-	1	Navigation Source (ADF #2)	-	
Navigation Source (VOR #1)	Discrete	1	211	4	4	0	-	1	Navigation Source (VOR #1)	-	
Navigation Source (VOR #2)	Discrete	1	211	5	5	0	-	1	Navigation Source (VOR #2)	-	
Preview Source (ADF #1)	Discrete	1	211	11	11	0	-	1	Preview Source (ADF #1)	-	
Preview Source (ADF #2)	Discrete	1	211	12	12	0	-	1	Preview Source (ADF #2)	-	



Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
Preview Source (FMS)	Discrete	1	211	8	8	0	-	1	Preview Source (FMS)	-	
Preview Source (VOR #1)	Discrete	1	211	9	9	0	-	1	Preview Source (VOR #1)	-	
Preview Source (VOR #2)	Discrete	1	211	10	10	0	-	1	Preview Source (VOR #2)	-	
Navigation Source SDI	Discrete	1	211	2	1	0	-	1	0 – PFD #1 1 – MFD #1 2 – PFD #2 3 – MFD #2		
NG LH NG	BNR	1	160-161	9-1	12-6	0	%rpm	0.005	-	-	
NG RH NG	BNR	1	163-164	9-1	12-6	0	%rpm	0.005	-	-	
NP LH NP	BNR	1	156-157	9-1	12-6	0	%rpm	0.005	-	-	
NP RH NP	BNR	1	158-159	9-1	12-6	0	%rpm	0.005	-	-	
OAT Outside Air Temperature #1	BNR	1	125-126	9-1	12-10	0	deg C	0.25	-	-	
OAT Outside Air Temperature #2	BNR	1	127-128	9-1	12-10	0	deg C	0.25	-	-	
PFD #1 Format - Map Display - Weather Radar Display - Traffic Display - HSI Display - Audio-Radio Display - Nav Log Display - Engine Display - Doors-Gear Display - Full Map Display - Full Engine Display	Discrete	1	189-190	8-1	12-5	0	-	1	0 – 0 1 – 1 2 – 2 3 – 3 4 – 4 5 – 5 6 – 6 7 – 7 8 – 8 9 – 9		Raw decimal and their corresponding output values have not been provided hence coded as shown.
PFD #1 Format Essential Mode	Discrete	1	190	2	2	0	-	1	Essential Mode	-	
PFD #2 Format - Map Display - Weather Radar Display - Traffic Display - HSI Display - Audio-Radio Display	Discrete	1	206-207	8-1	12-5	0	-	1	0 – 0 1 – 1 2 – 2 3 – 3 4 – 4 5 – 5		Raw decimal and their corresponding output values have not been provided hence coded as shown.

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
- Nav Log Display - Engine Display - Doors-Gear Display - Full Map Display - Full Engine Display									6 – 6 7 – 7 8 – 8 9 – 9		
PFD #2 Format Essential Mode	Discrete	1	207	2	2	0	-	1	Essential Mode	-	
Pitch Attitude Pitch Attitude #1	BNR	0.25	10-11, 74-75, 138-139, 202-203	9-1	12-9	0	deg	0.0055	-	-	
Pitch Attitude Pitch Attitude #2	BNR	0.25	12-13, 76-77, 140-141, 204-205	9-1	12-9	0	deg	0.0055	-	-	
Pitch/Elevator Trim Position LH Pitch/Elevator Trim Surface Pos	BNR	1	216-217	9-1	12-10	0	deg	0.088	-	-	
Pitch/Elevator Trim Position RH Pitch/Elevator Trim Surface Pos	BNR	1	218-219	9-1	12-10	0	deg	0.088	-	-	
Pressure Altitude Pressure Altitude #1	BNR	1	117-118	9-1	12-4	0	feet	1	-	-	
Pressure Altitude Pressure Altitude #2	BNR	1	119-120	9-1	12-4	0	feet	1	-	-	
Propeller Oil Press LH Propeller Oil Pressure	BNR	1	88-89	8-1	12-9	0	psi	0.1	-	-	
Propeller Oil Press RH Propeller Oil Pressure	BNR	1	90-91	8-1	12-9	0	psi	0.1	-	-	
Propeller Oil Temp LH Propeller Oil Temperature	BNR	1	80-81	9-1	12-9	0	deg C	0.1	-	-	
Propeller Oil Temp RH Propeller Oil Temperature	BNR	1	82-83	9-1	12-9	0	deg C	0.1	-	-	
Radio Altimeter Radio Altitude	BNR	1	105-106	8-1	12-5	0	feet	0.125	-	-	
RH Aileron Pos RH Roll Axis Control Surface Pos	BNR	0.25	7, 71, 135, 199	12	1	0	deg	-23.527, 0.01	-	-	
RH Elevator Pos RH Pitch Control Surface/Elevator Pos	BNR	0.25	9, 73, 137, 201	12	1	0	deg	-40, 0.0195	-	-	
RHReverse Thrust Reverser Position RH	Discrete	1	248	5	5	0	-	1	Open	Ground	
Roll Attitude Roll Attitude #1	BNR	0.5	18-19, 146-147	9-1	12-9	0	deg	0.0055	-	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
Roll Attitude Roll Attitude #2	BNR	0.5	20-21, 148-149	9-1	12-9	0	deg	0.0055	-	-	
Roll/Aileron Trim Position LH Roll/Aileron Trim Surface Pos	BNR	1	220-221	9-1	12-10	0	deg	0.088	-	-	
Roll/Aileron Trim Position RH Roll/Aileron Trim Surface Pos	BNR	1	222-223	9-1	12-10	0	deg	0.088	-	-	
Rudder Pos Yaw Axis Control Surface Pos	BNR	0.25	5, 69, 133, 197	12	1	0	deg	-11.26, 0.0055	-	-	
Rudder Trim Position Rudder/Yaw Trim Surface Pos	BNR	1	227-228	9-1	12-10	0	deg	0.088	-	-	
Selected Al. Selected Altitude	BNR	1	49-50	9-1	12-5	0	feet	1	-	-	
Selected Course Selected Course	BNR	1	45-46	9-1	12-9	0	deg	0.0434	-	-	
Selected Heading Selected Heading	BNR	1	47-48	9-1	12-9	0	deg	0.0434	-	-	
Selected Speed Selected Speed	BNR	1	51-52	9-1	12-5	0	knots	0.25	-	-	
Selected VS Selected Vertical Speed	BNR	1	53-54	9-1	12-9	0	fpm	4	-	-	
SYNC Sync Word	BNR	1	1	12	1	0	oct	1	-	-	
TAWS Caution Obstacle	Discrete	1	39	2	2	0	-	1	Caution Obstacle	-	
TAWS Caution Terrain	Discrete	1	39	1	1	0	-	1	Caution Terrain	-	
TAWS Don't Sink	Discrete	1	40	7	7	0	-	1	Don't Sink	-	
TAWS Fifty	Discrete	1	41	1	1	0	-	1	Fifty	-	
TAWS Five Hundred	Discrete	1	41	4	4	0	-	1	Five Hundred	-	
TAWS Forty	Discrete	1	42	12	12	0	-	1	Forty	-	
TAWS Glideslope	Discrete	1	40	11	11	0	-	1	Glideslope	-	
TAWS Glideslope Cancel	Discrete	1	38	3	3	0	-	1	Glideslope Cancel	-	
TAWS Minimums	Discrete	1	42	8	8	0	-	1	Minimums	-	
TAWS Minimums/Minimums	Discrete	1	40	12	12	0	-	1	Minimums/Minimums	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
TAWS Mode 1 Pull Up	Discrete	1	39	6	6	0	-	1	Mode 1 Pull Up	-	
TAWS Mode 1 Sinkrate	Discrete	1	39	5	5	0	-	1	Mode 1 Sinkrate	-	
TAWS Mode 2 Pull Up	Discrete	1	39	8	8	0	-	1	Mode 2 Pull Up	-	
TAWS Mode 2 Terrain	Discrete	1	39	7	7	0	-	1	Mode 2 Terrain	-	
TAWS Mode 3 Visual	Discrete	1	42	3	3	0	-	1	Mode 3 Visual	-	
TAWS Mode 4 Visual	Discrete	1	42	4	4	0	-	1	Mode 4 Visual	-	
TAWS Mode 5 Visual	Discrete	1	42	5	5	0	-	1	Mode 5 Visual	-	
TAWS Modes 1-4 Inop	Discrete	1	38	9	9	0	-	1	Modes 1-4 Inop	-	
TAWS Modes 5 Inop	Discrete	1	38	10	10	0	-	1	Modes 5 Inop	-	
TAWS Modes 6 Inop	Discrete	1	38	11	11	0	-	1	Modes 6 Inop	-	
TAWS Obstacle Awareness Caution	Discrete	1	44	3	3	0	-	1	Obstacle Awareness Caution	-	
TAWS Obstacle Awareness Warning	Discrete	1	44	4	4	0	-	1	Obstacle Awareness Warning	-	
TAWS Obstacle/Obstacle	Discrete	1	39	3	3	0	-	1	Obstacle/ Obstacle	-	
TAWS One Hundred	Discrete	1	41	2	2	0	-	1	One Hundred	-	
TAWS Pull Up	Discrete	1	40	5	5	0	-	1	Pull Up	-	
TAWS RALT Fail	Discrete	1	38	6	6	0	-	1	RALT Fail	-	
TAWS Sinkrate	Discrete	1	40	4	4	0	-	1	Sinkrate	-	
TAWS TAWS Enable MFD Copilot Side	Discrete	1	37	8	8	0	-	1	TAWS Enable MFD	-	
TAWS TAWS Enable MFD Pilot Side	Discrete	1	37	6	6	0	-	1	TAWS Enable MFD	-	



Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
TAWS TAWS Enable PFD Copilot Side	Discrete	1	37	7	7	0	-	1	TAWS Enable PFD	-	
TAWS TAWS Enable PFD Pilot Side	Discrete	1	37	5	5	0	-	1	TAWS Enable PFD	-	
TAWS TAWS Inhibit	Discrete	1	38	5	5	0	-	1	TAWS Inhibit	-	
TAWS TAWS Pop-Up Enable	Discrete	1	40	3	3	0	-	1	Pop-Up Enable	-	
TAWS Ten	Discrete	1	42	9	9	0	-	1	Ten	-	
TAWS Terrain	Discrete	1	40	6	6	0	-	1	Terrain	-	
TAWS Terrain Awareness Caution	Discrete	1	41-42	6-6	6-6	0	-	1	Terrain Awareness Caution	-	
TAWS Terrain Awareness Warning	Discrete	1	41-42	7-7	7-7	0	-	1	Terrain Awareness Warning	-	
TAWS Terrain Awareness Warning	Discrete	1	41-42	7-7	7-7	0	-	1	Terrain Awareness Warning	-	
TAWS Thirty	Discrete	1	42	11	11	0	-	1	Thirty	-	
TAWS Three Hundred	Discrete	1	41	3	3	0	-	1	Three Hundred	-	
TAWS Too Low Flaps	Discrete	1	40	9	9	0	-	1	Too Low Flaps	-	
TAWS Too Low Gear	Discrete	1	40	8	8	0	-	1	Too Low Gear	-	
TAWS Too Low Terrain	Discrete	1	40	10	10	0	-	1	Too Low Terrain	-	
TAWS Twenty	Discrete	1	42	10	10	0	-	1	Twenty	-	
TCAS Advisory Rate To Maintain	BNR	1	168	9	3	0	fpm	100	-	-	Bit 11 : 100 ft/min Bit 12 : 200 ft/min Bit 13 : 400 ft/min Bit 14 : 800 ft/min Bit 15 : 1600 ft/min Bit 16 : 3200 ft/min Bit 17 : Sign
TCAS Combined Control	Discrete	1	168	12	10	0	-	1	0 – No Advisory 1 – Clear of Conflict		

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
											2 – Drop Track 3 – Altitude Lost 4 – Up Advisory Corrective 5 – Down Advisory Corrective 6 – Preventive 7 – Not Used
TCAS Down Advisory	Discrete	1	167	9	7	0	-	1			0 – No Down Advisory 1 – Descend 2 – Do Not Climb 3 – No Climb > 500 FPM 4 – No Climb > 1000 FPM 5 – No Climb > 2000 FPM 6 – Not Used 7 – Not Used
TCAS Up Advisory	Discrete	1	167	6	4	0	-	1			0 – No Up Advisory 1 – Climb 2 – Do Not Descend 3 – No Descend > 500 4 – No Descend > 1000 5 – No Descend > 2000 6 – Not Used 7 – Not Used
TCAS Vertical Control	Discrete	1	167	3	1	0	-	1			0 – Advisory is None of the Following Types 1 – Crossing 2 – Reversal 3 – Increase 4 – Maintain 5 – Not Used 6 – Not Used 7 – Not Used
Torque LH Torque Eng	BNR	1	243-244	9-1	12-6	0	ft-lb	0.005	-	-	
Torque RH Torque Eng	BNR	1	245-246	9-1	12-6	0	ft-lb	0.005	-	-	
UTC GMT Hours	BNR	1	104	8	4	0	hour	-	-	-	Raw Decimal
UTC GMT Minutes	BNR	1	103-104	3-1	12-10	0	minute	-	-	-	Raw Decimal

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
UTC GMT Seconds	BNR	1	103	8	4	0	second	-	-	-	Raw Decimal
Vertical Acc Vertical Acceleration	BNR	0.125	2, 34, 66, 98, 130, 162, 194, 226	12	1	0	G	-3.3756, 0.0023	-	-	
VOR VOR Frequency #1 (Hundreths)	BCD	1	29	10	7	0	MHz	1	-	-	
VOR VOR Frequency #1 (Tens)	BCD	1	28	9	7	0	MHz	1	-	-	
VOR VOR Frequency #1 (Tenths)	BCD	1	28-29	2-1	12-11	0	MHz	1	-	-	
VOR VOR Frequency #1 (Units)	BCD	1	28	6	3	0	MHz	1	-	-	
VOR VOR Frequency #2 (Hundreths)	BCD	1	31	10	7	0	MHz	1	-	-	
VOR VOR Frequency #2 (Tens)	BCD	1	30	9	7	0	MHz	1	-	-	
VOR VOR Frequency #2 (Tenths)	BCD	1	30-31	2-1	12-11	0	MHz	1	-	-	
VOR VOR Frequency #2 (Units)	BCD	1	30	6	3	0	MHz	1	-	-	
Warning Messages Auto Feather	Discrete	1	174	4	4	0	-	1	Auto Feather	-	
Warning Messages Cabin Alt Warning	Discrete	1	176	4	4	0	-	1	Cabin Alt Warning	-	
Warning Messages Cargo Door	Discrete	1	176	8	8	0	-	1	Cargo Door	-	
Warning Messages Cargo Fire	Discrete	1	172	4	4	0	-	1	Cargo Fire	-	
Warning Messages Flap Asym	Discrete	1	172	5	5	0	-	1	Flap Asym	-	
Warning Messages Lav Fire	Discrete	1	170	4	4	0	-	1	Lav Fire	-	
Warning Messages LH AFT Brk OH	Discrete	1	173	1	1	0	-	1	LH AFT Brk OH	-	
Warning Messages LH AFT Door	Discrete	1	174	10	10	0	-	1	LH AFT Door	-	
Warning Messages LH Bleed Hot	Discrete	1	170	7	7	0	-	1	LH Bleed Hot	-	
Warning Messages LH DC Gen Hot Warning	Discrete	1	174	12	12	0	-	1	LH DC Gen Hot Warning	-	



Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
Warning Messages LH DC Gen Warning	Discrete	1	174	11	11	0	-	1	LH DC Gen Warning	-	
Warning Messages LH Engine Fire Warning	Discrete	1	170	6	6	0	-	1	LH Engine Fire Warning	-	
Warning Messages LH Engine Oil Press Warning	Discrete	1	174	5	5	0	-	1	LH Engine Oil Press Warning	-	
Warning Messages LH FWD Brk OH	Discrete	1	174	7	7	0	-	1	LH FWD Brk OH	-	
Warning Messages LH FWD Door	Discrete	1	174	9	9	0	-	1	LH FWD Door	-	
Warning Messages LH Landing Gear Down	Discrete	1	172	9	9	0	-	1	LH Landing Gear Down	-	
Warning Messages LH Landing Gear Up	Discrete	1	170	10	10	0	-	1	LH Landing Gear Up	-	
Warning Messages LH Propeller Oil Press Warning	Discrete	1	174	6	6	0	-	1	LH Propeller Oil Press Warning	-	
Warning Messages Nose Landing Gear Down	Discrete	1	170	8	8	0	-	1	Nose Landing Gear Down	-	
Warning Messages Nose Landing Gear Up	Discrete	1	172	11	11	0	-	1	Nose Landing Gear Up	-	
Warning Messages Overspeed Warning	Discrete	1	172	3	3	0	-	1	Overspeed Warning	-	
Warning Messages Prop Brk	Discrete	1	170	5	5	0	-	1	Prop Brk	-	
Warning Messages RH AFT Brk OH	Discrete	1	175	1	1	0	-	1	RH AFT Brk OH	-	
Warning Messages RH AFT Door	Discrete	1	176	10	10	0	-	1	RH AFT Door	-	
Warning Messages RH Bleed Hot	Discrete	1	172	7	7	0	-	1	RH Bleed Hot	-	
Warning Messages RH DC Gen Hot Warning	Discrete	1	176	12	12	0	-	1	RH DC Gen Hot Warning	-	

Dataframe Layout Document – Recorded Parameters



Parameter Mnemonic & Name	Parameter Type	Frequency	Word Location	A717 MSB	A717 LSB	Subframe	Units	Resolution	True State	False State	Notes
Warning Messages RH DC Gen Warning	Discrete	1	176	11	11	0	-	1	RH DC Gen Warning	-	
Warning Messages RH Engine Fire Warning	Discrete	1	172	6	6	0	-	1	RH Engine Fire Warning	-	
Warning Messages RH Engine Oil Press Warning	Discrete	1	176	5	5	0	-	1	RH Engine Oil Press Warning	-	
Warning Messages RH FWD Brk OH	Discrete	1	176	7	7	0	-	1	RH FWD Brk OH	-	
Warning Messages RH FWD Door	Discrete	1	176	9	9	0	-	1	RH FWD Door	-	
Warning Messages RH Landing Gear Down	Discrete	1	172	9	9	0	-	1	RH Landing Gear Down	-	
Warning Messages RH Landing Gear Up	Discrete	1	172	10	10	0	-	1	RH Landing Gear Up	-	
Warning Messages RH Propeller Oil Press Warning	Discrete	1	176	6	6	0	-	1	RH Propeller Oil Press Warning	-	
Warning Messages Stall	Discrete	1	174	8	8	0	-	1	Stall	-	
Warning Messages Wing OH	Discrete	1	170	3	3	0	-	1	Wing OH	-	
Wind Direction Wind Direction	BNR	1	231	9	1	0	deg	0.7	-	-	
Wind Speed Wind Speed	BNR	1	229	8	1	0	deg	1	-	-	
WOWLH Air-Ground Status LH	Discrete	1	249	2	2	0	-	1	Open	Ground	
WOWNOSE Air-Ground Status Nose	Discrete	1	248	1	1	0	-	1	Open	Ground	
WOWRH Air-Ground Status RH	Discrete	1	248	3	3	0	-	1	Open	Ground	